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Windows PowerShell
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PS C:\Users\Prateek> ubuntu
prateek@prateek:~$ nano metrics_threads_ipc.py
prateek@prateek:~$ python3 metrics_threads_ipc.py

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ROUND ROBIN SCHEDULING
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Time Quantum: 2

Gantt Chart:
| P1 [0-2] | P2 [2-4] | P3 [4-6] | P1 [6-8] | P2 [8-9] | P3 [9-11] | P4 [11-13] | P1 [13-14] |


Execution Intervals:
P1: Start = 0, End = 2
P2: Start = 2, End = 4
P3: Start = 4, End = 6
P1: Start = 6, End = 8
P2: Start = 8, End = 9
P3: Start = 9, End = 11
P4: Start = 11, End = 13
P1: Start = 13, End = 14

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SCHEDULING METRICS
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PID     AT      BT      CT      TAT     WT      RT
P1      0       5       14      14      9       0
P2      1       3       9       8       5       1
P3      2       4       11      9       5       2
P4      8       2       13      5       3       3

Average Turnaround Time: 9.0
Average Waiting Time: 5.5
Average Response Time: 1.5

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THREADING
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Thread-1: Thread ID = 140325812942592
Thread-1: completed work 1
Thread-2: Thread ID = 140325804550144
prateek@prateek:~$
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Thread-2: completed work 1
Thread-2: completed work 2
Thread-1: completed work 2
Thread-2: completed work 3
Thread-1: completed work 3
Thread-1: work completed.
Thread-2: work completed.
Both threads completed successfully.

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INTER-PROCESS COMMUNICATION (IPC)
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Sender Process ID: 1209
Transmitted message: Hello from sender process
Receiver Process ID: 1204
Received message: Hello from sender process

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EXPLANATION
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1. Scheduling Results:
   Round Robin gives each process a fixed time quantum.
   Processes are executed in FIFO ready-queue order.
   CPU idle time is shown whenever no process is available.

2. Benefits of Threads:
   - Threads are lightweight.
   - They share memory within a process.
   - They can perform multiple tasks concurrently.
   - They are useful for responsive applications.

3. IPC Method Used:
   multiprocessing.Pipe() was used for communication.

4. Limitation of Pipe:
   A Pipe is mainly suitable for communication between
   a small number of processes. It is not as convenient
   as a Queue for communication involving many processes.

5. Conclusion:
   Scheduling metrics, Round Robin execution,
   multithreading and inter-process communication
   were successfully demonstrated.

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ASSIGNMENT COMPLETED SUCCESSFULLY
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